NUCES FAST, Islamabad

Deep Learning

Assignment No.01

Instructors

Mahnoor Tariq & Dr. Qurat Ul Ain

September 10, 2025

Announcement

This is to announce that your assignment on CNN models is now available. Please make sure to complete the task and follow all given instructions. Submit your assignment before the due date.

Due Date: 27th September

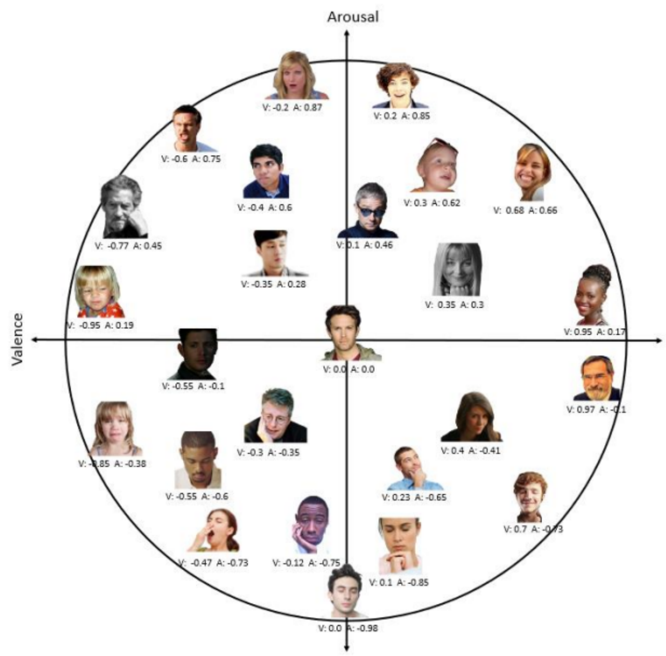
1 Background

AFFECT is a psychological term used to describe the outward expression of emotion and feelings. Affective computing seeks to develop systems and devices that can recognize, interpret, and simulate human affects through various channels such as face, voice, and biological signals. Face and facial expressions are undoubtedly one of the most important nonverbal channels used by the human being to convey internal emotion.

Some Sample Facial Expressions

Dimensional model of affect is related with Quantification of affect into continuous values of valence and arousal. Valence refers to how positive or negative an event is, and arousal reflects whether an event is exciting/agitating or calm/soothing" Valence: Whether expression is positive or negative
Arousal: Whether event is exciting/agitating or calm/soothing
In the given dataset of face Images, following is provided

- Location of the faces in the images
- Location of the 68 facial landmarks
- Eight emotion and non-emotion categorical labels (0: Neutral, 1: Happy, 2: Sad, 3: Surprise, 4: Fear, 5: Disgust, 6: Anger, 7: Contempt)
- Valence and arousal values of the facial expressions in continuous domain, and are provided as floating point numbers in the interval [-1,+1].



Sample images in Valence Arousal circumplex with their corresponding Valence and Arousal values (V: Valence, A: Arousal).

The images are cropped and resized to 224 x 224 pixels (RGB color). The 8 expression labels and the Arosal/Valence values as well as the facial landmark points of the training and test set are in the database. You can split the training dataset in to train and validation.

2 Notes for Dataset

- The x and y coordination and the annotations are stored in .npy files. Use Python numpy to read the files separated with a semi-colon.
- Expression: expression ID of the face (0: Neutral, 1: Happy, 2: Sad, 3: Surprise, 4: Fear, 5: Disgust, 6: Anger, 7: Contempt)
- Valence: valence value of the expression in interval [-1,+1] (for Uncertain and Noface categories the value is -2) Continuous values from -1 (negative/unpleasant) to +1 (positive/pleasant)
- Arousal: arousal value of the expression in interval [-1,+1] (for Uncertain and No-face categories the value is -2) Continuous values from -1 (Tired) to +1 (Active)

Evaluation Metrics

Report your results in following evaluation metrics

For categorical classification

Name	Metric	Description
Accuracy	Accuracy	
F1-Score	F1-Score	
Cohens Kappa	Kappa	Statistical measure of agreement
Krippendorfs Alpha	Alpha	Statistical measure of agreement
Area Under Curve	AUC	Area under ROC curve
AUC – Precision Recall	AUC-PR	Area under Precision-Recall curve

For Continuous domain

Name	Metric	Description
Root Mean Square Error	RMSE	
Correlation	CORR	
Sign Agreement Metric	SAGR	Penalizes incorrect sign alongside deviation from value (e.g., For ground truth +0.3, +0.7 is better prediction than -0.1)
Concordance Correlation Coefficient	CCC	Combines the Pearsons correlation coefficient (CC) with the square difference between the means of two compared time series

3 Task

Use appropriate CNN architecture for recognition of Facial Expression, and computing Valence and Arousal, for the given dataset. Please use at least two CNN baselines (VGG, ResNet, Inception, Xception, MobileNet, EfficientNet, SE Net, DenseNet or any other ... etc) and give the comparison of their results.

Deliverables

Implement your task using IDE, Python notebook or Google Colab. You may download PyCharm.

1. The code files (.ipynb) to be uploaded on your GitHub repository and the link to be uploaded on LMS. Do not share links of Google Colaboratory, rather upload on GitHub as Python Notebook.
    - The Python Notebook should also demonstrate:
        - Neatly written, documented and modular code

    Strongly encouraged to use Best Practices and Modular implementation of Image classification pipeline.

    Links (wrapped automatically):

    - `https://github.com/GoogleCloudPlatform/keras-idiomatic-programmer`
    - `https://github.com/GoogleCloudPlatform/keras-idiomatic-programmer/tree/master/handbooks`
    - keras-idiomatic-programmer (GitHub)
2. Short Report: Should contain the following:
    - Network details: architecture, parameters, training setting, etc.
    - Dataset splits
    - Training graphs
    - Performance measures
    - Performance comparison of CNN architectures (accuracy & time)

    Naming Convention: `<your FastID>_<your name>_A1-CS452.pdf`

Grading Rubric

The assignment is of **100 pts** in total.

The breakdown is given below.

Short Report: 50 points. Breakdown given below

- Network details which include architecture, parameters, training setting, batch size etc. A note on rationale of choosing the baseline. [**10pt**]
- If Multiple baselines are chosen, a comparison on performance of these baselines. [**10 pt**]
- Details of any Transfer learning used [**5 pt**]
- Training graphs, (showing decreasing loss and increasing accuracy with number of epochs) [**5 pt**]
- Performance measures and a discussion on continuous domain evaluation metrics (RMSE, CORR, SAGR, CCC) — what is the rationale of using each one and in the present scenario, which should be most suited if we want to design a system which should work in the wild. [**15 pt**]
- A few of the correctly and incorrectly classified images. [**4 pt**]
- Use naming convention while submitting the PDF file [**1 pt**]

Code: 50 points. Breakdown given below.

- Code is properly **documented, modular and understandable** [**10 points**]
- Modular implementation of pipeline. Use functions and object-oriented implementation, as illustrated in keras-idiomatic-programmer guide. If you are using Pytorch, object-oriented implementation counts for this. [**10 points**]
- Use of appropriate data augmentation, on the fly or off-line [**6 points**]
- Use of multiple CNN architectures as baseline (use at least two) and appropriately shown comparison of Quantitative performance measures [**20 points**]

  (If quantitative performance measures are too low, then points will be deduced accordingly).
- Qualitative results i.e. a few of the correctly and incorrectly classified images [**4 points**]

Extra Credit:

Develop your custom CNN architecture and it performs better than your chosen baseline architecture. Include a discussion section in report illustrating the rationale of proposed architecture of your custom CNN. [**5 pts**]

Dataset availability link:

https://drive.google.com/file/d/1PdQBPUuHSBfEtK30cV2IIpRFBJ4-8B78/view?usp=sharing

Use your FAST email ID to download it. Do not send access requests from your private email IDs.